OPEN ANALOG IC DESIGN

SKY130A

PROJECT 2-STAGE OTA

Two-Stage OTA Design in SKY130A

Specification-Driven Analog Design from Ideal Architecture to Transistor Level

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1 Project Overview

This project develops a classical two-stage CMOS OTA using the SKY130A open-source PDK.

The intended application is a low-power analog block representative of PMIC and voltage-reference circuits, where DC gain, low current consumption, stability and supply rejection are prioritized over maximum speed.

The design follows a specification-driven methodology. First-order analytical calculations are used to establish a coherent initial design before transistor sizing. An ideal architectural model is then used to study the fundamental two-stage behavior and compensation strategy before implementing the complete transistor-level circuit.

Transistor sizing reuses the SKY130A device characterization obtained in the preceding `project_carac_sky130A` project, including g_m/I_D, I_D/W, intrinsic gain, threshold-voltage and saturation-voltage data.

The design flow used throughout the project is:

1. define the OTA specifications,
2. select the topology and bias-current architecture,
3. derive a first-order analytical design,
4. study the ideal two-stage architecture and compensation,
5. size the SKY130A transistors using the characterized device data,
6. verify and iterate at transistor level,
7. characterize the final OTA over the relevant operating conditions.

2 Specifications and Architecture

2.1 Initial Design Specifications

The OTA is intended for a low-power PMIC or voltage-reference application. The design therefore prioritizes DC gain, low current consumption, stability and supply rejection over maximum speed.

The initial design targets are summarized in Table 1.

Table 1 Initial OTA design specifications.

Parameter	Initial target
Supply voltage	1.8 V
DC gain	$\geq$ 80 dB
GBW	$\geq$ 1 MHz
Phase margin	$\geq 60^\circ$
Load capacitance	2 pF nominal, 4 pF maximum
Quiescent current	Minimize within the other constraints
PSRR	High priority, characterized versus frequency
Slew rate	Secondary design objective

2.2 Selected Topology

A classical two-stage Miller-compensated CMOS OTA is selected.

The first stage uses a PMOS differential pair with an NMOS current-mirror active load. Its single-ended output node, `out_s1`, drives the second stage.

The second stage consists of the NMOS common-source transistor `NLOAD_2`, biased by the PMOS current source `PTAIL_2`. The OTA output is denoted `Vout`.

The compensation network is connected between `out_s1` and `Vout`. It contains the Miller capacitor `CMILLER` and, when required, the series resistor `RMILLER`.

2.3 Device Naming

The naming convention shown in Figure 1 is used throughout the analytical design and simulation results.

- `PTAIL_0`: diode-connected PMOS receiving the reference bias current and generating the PMOS mirror bias voltage,
- `PTAIL_1`: PMOS current source biasing the first stage,
- `PDIFF_P`, `PDIFF_N`: PMOS input differential pair,
- `NLOAD_0`, `NLOAD_1`: NMOS current-mirror active load,
- `out_s1`: first-stage output and second-stage input,
- `PTAIL_2`: PMOS current source biasing the second stage,
- `NLOAD_2`: NMOS common-source transistor of the second gain stage,
- `Vout`: OTA output,

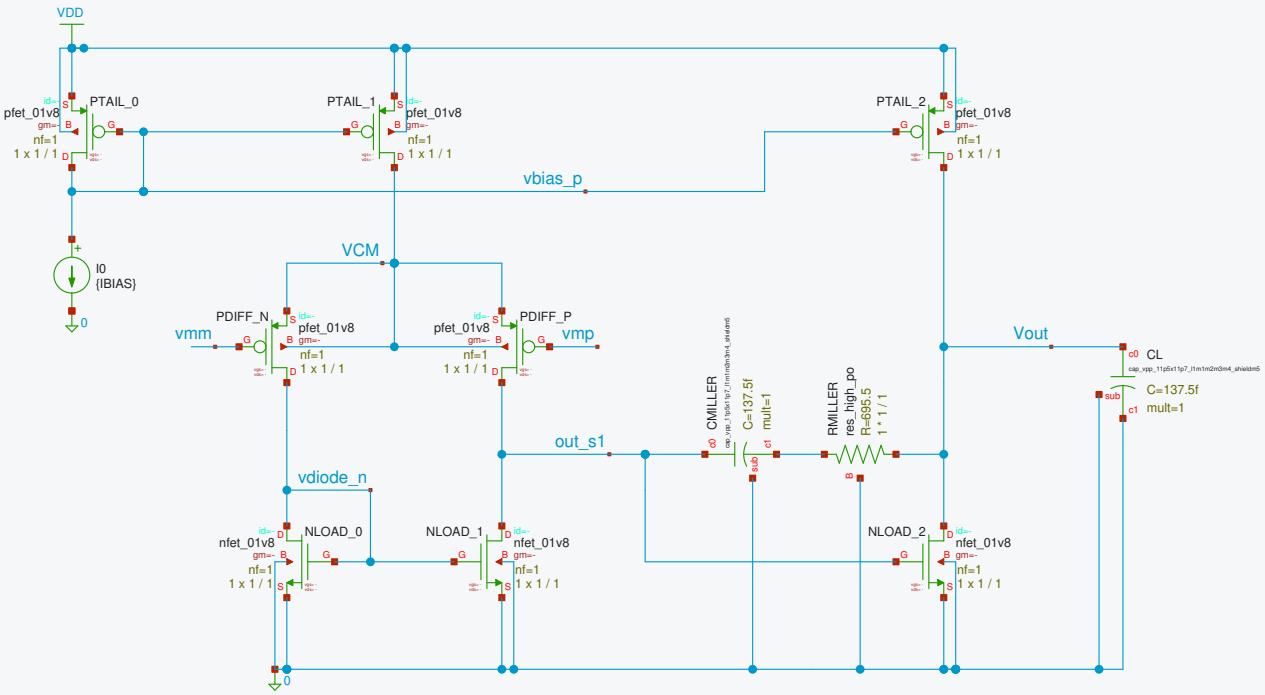


Figure 1 Selected two-stage OTA topology and device naming convention.

- `CMILLER`: Miller compensation capacitor,
- `RMILLER`: optional series compensation resistor.

2.4 Bias-Current Architecture

The diode-connected device `PTAIL_0` defines the reference current unit I.

The first-stage current is initially defined as

$$I_{\mathrm{PTAIL_1}} = I,$$

which gives, at zero differential input,

$$I_{\mathrm{PDIFF_P}} = I_{\mathrm{PDIFF_N}} = \frac{I}{2}.$$

The second-stage current is expressed as

$$I_{\mathrm{PTAIL_2}} = NI,$$

where the mirror ratio N is derived from the required second-stage transconductance and stability constraints in the first-order design.

3 First-Order Design

The OTA is first sized analytically to obtain a coherent initial operating point before transistor-level optimization. These calculations are intended as first-order design guidance rather than exact predictions of the final BSIM-based circuit.

3.1 Bias-Current Selection

The diode-connected PMOS `PTAIL_0` defines the reference bias current

$$I = 500\ \text{nA}.$$

The first-stage PMOS current source is initially mirrored with a ratio of one:

$$I_{\text{PTAIL_1}} = I = 500\ \text{nA}.$$

At zero differential input, the differential-pair current therefore splits equally:

$$I_{\text{PDIFF_P}} = I_{\text{PDIFF_N}} = \frac{I}{2} = 250\ \text{nA}.$$

The second-stage bias current is expressed as a multiple of the reference current:

$$I_{\text{PTAIL_2}} = NI.$$

The value of N is derived from the required second-stage transconductance and stability constraints.

3.2 Input-Stage Transconductance

For the low-power input differential pair, an initial transconductance efficiency of

$$\frac{g_m}{I_D} \approx 20\ \text{V}^{-1}$$

is selected.

With

$$I_D = 250\ \text{nA},$$

the expected input-device transconductance is

$$g_{m1} = \frac{g_m}{I_D} I_D = 20 \times 250\ \text{nA} = 5\ \mu\text{S}.$$

3.3 Miller Compensation Capacitor

For a classical Miller-compensated two-stage OTA, the unity-gain bandwidth is approximated by

$$GBW \approx \frac{g_{m1}}{2\pi C_{\text{MILLER}}}.$$

Using the target

$$GBW = 1\ \text{MHz},$$

the required compensation capacitance is

$$C_{\text{MILLER}} = \frac{5\ \mu\text{S}}{2\pi \times 1\ \text{MHz}} \approx 0.80\ \text{pF}.$$

The initial compensation value is therefore

$$\boxed{C_{\text{MILLER}} \approx 0.8\ \text{pF}}.$$

A SKY130A MIM capacitor is used for the physical implementation of this component.

3.4 Second-Stage Transconductance

The second-stage transconductance must place the non-dominant output pole sufficiently above the unity-gain frequency.

Using the first-order approximation

$$\omega_{p2} \approx \frac{g_{m2}}{C_L}$$

and targeting

$$\omega_{p2} \gtrsim 2.2\,\omega_u$$

for approximately 60° phase margin gives

$$g_{m2} \gtrsim 2.2\,g_{m1}\frac{C_L}{C_{\text{MILLER}}}.$$

For the worst-case initial load

$$C_L = 4\ \text{pF},$$

the required second-stage transconductance is approximately

$$g_{m2} \gtrsim 2.2 \times 5\ \mu\text{S} \times \frac{4\ \text{pF}}{0.8\ \text{pF}} = 55\ \mu\text{S}.$$

The initial target is therefore

$$\boxed{g_{m2} \approx 55\text{–}60\ \mu\text{S}}.$$

3.5 Second-Stage Bias Ratio

An initial second-stage current of

$$4I = 2\ \mu\text{A}$$

would require

$$\frac{g_m}{I_D} \approx \frac{55\ \mu\text{S}}{2\ \mu\text{A}} = 27.5\ \text{V}^{-1}.$$

This operating point is unnecessarily aggressive for the initial design.

Increasing the mirror ratio to

$$N = 6$$

gives

$$I_{\text{PTAIL_2}} = 6I = 3\ \mu\text{A}.$$

The corresponding required transconductance efficiency becomes

$$\frac{g_m}{I_D} \approx \frac{55\ \mu\text{S}}{3\ \mu\text{A}} \approx 18.3\ \text{V}^{-1},$$

which is a more suitable initial operating region.

The initial bias-current ratio is therefore selected as

$$\boxed{I_{\text{PTAIL_0}} : I_{\text{PTAIL_1}} : I_{\text{PTAIL_2}} = 1 : 1 : 6}.$$

3.6 First-Order Slew-Rate Estimate

Although slew rate is not a primary design objective, it provides a useful large-signal sanity check.

Assuming that the first-stage bias current limits the charging or discharging of the Miller capacitor,

$$SR \approx \frac{I_{\text{PTAIL_1}}}{C_{\text{MILLER}}}.$$

Therefore,

$$SR \approx \frac{500\ \text{nA}}{0.8\ \text{pF}} \approx 0.63\ \text{V}/\mu\text{s}.$$

This value is considered sufficient as an initial estimate for the intended low-power precision application.

3.7 Initial Transistor Sizing

The transistor dimensions are obtained from the SKY130A device characterization performed in the preceding project.

For each device group, the drain current, target g_m/I_D and channel length are first selected from the circuit-level requirements. The characterized current density I_D/W is then used to determine the required width.

3.7.1 Input Differential Pair

The PMOS differential pair carries

$$I_D = 250\ \text{nA}$$

per device and targets

$$\frac{g_m}{I_D} \approx 20\ \text{V}^{-1}.$$

A channel-length comparison showed that increasing the length from 1 μm to 2 μm strongly increased device area for comparatively limited circuit-level benefit.

The initial geometry is therefore selected as

$$\boxed{W_{\text{PDIFF}} = 89\ \mu\text{m}, \qquad L_{\text{PDIFF}} = 1\ \mu\text{m}}.$$

At this operating point, the characterized intrinsic gain is approximately

$$\frac{g_m}{g_{ds}} \approx 478.$$

3.7.2 First-Stage NMOS Mirror Load

The two NMOS mirror devices also carry

$$I_D = 250\ \text{nA}.$$

A preliminary target of $g_m/I_D \approx 15\ \text{V}^{-1}$ resulted in a calculated width below the minimum practical device width.

The width was therefore fixed to

$$W = 0.42\ \mu\text{m},$$

and the corresponding operating point was re-evaluated from the characterized data.

For $L = 1\ \mu$m, the resulting values are approximately

$$\frac{g_m}{I_D} \approx 19.5\ \text{V}^{-1}$$

and

$$\frac{g_m}{g_{ds}} \approx 210.$$

The initial geometry is therefore

$$\boxed{W_{\text{NLOAD0,1}} = 0.42\ \mu\text{m}, \qquad L_{\text{NLOAD0,1}} = 1\ \mu\text{m}}.$$

3.7.3 Second-Stage Gain Device

The NMOS second-stage gain transistor carries

$$I_D = 3\ \mu\text{A}$$

and targets

$$\frac{g_m}{I_D} \approx 18.3\ \text{V}^{-1}.$$

The characterized data showed little intrinsic-gain improvement between $L = 1\ \mu$m and $L = 2\ \mu$m, while the required width approximately doubled.

The selected initial geometry is therefore

$$\boxed{W_{\text{NLOAD2}} = 3.58\ \mu\text{m}, \qquad L_{\text{NLOAD2}} = 1\ \mu\text{m}}.$$

3.7.4 PMOS Bias Mirror

The PMOS bias devices are operated around

$$\frac{g_m}{I_D} \approx 15\ \text{V}^{-1}.$$

For the reference current of 500 nA, the selected geometry is

$$W_{\text{PTAIL0}} = W_{\text{PTAIL1}} = 1.76\ \mu\text{m}, \qquad L = 1\ \mu\text{m}.$$

The second-stage current source uses the same operating point and a $6:1$ mirror ratio:

$$W_{\text{PTAIL2}} = 6 \times 1.76 = 10.56\ \mu\text{m}.$$

3.7.5 Initial Sizing Summary

Table 2 Initial transistor sizing obtained from first-order design and SKY130A characterized device data.

Device	I_D	g_m/I_D	W (μm)	L (μm)
`PTAIL_0`	500 nA	14.9	1.76	1.0
`PTAIL_1`	500 nA	14.9	1.76	1.0
`PDIFF_N/P`	250 nA	20.0	89.0	1.0
`NLOAD_0/1`	250 nA	19.5	0.42	1.0
`PTAIL_2`	3 μA	14.9	10.56	1.0
`NLOAD_2`	3 μA	18.2	3.58	1.0

3.8 First-Order Gain Estimate

Using the characterized output conductances at the selected operating points, the first-stage gain is estimated as

$$A_{v1} \approx g_{m1}\left(r_{o,\text{PDIFF}} \parallel r_{o,\text{NLOAD1}}\right) \approx 43.5\ \text{dB}.$$

For the second stage,

$$A_{v2} \approx g_{m2} \left(r_{o,\mathrm{NLOAD2}} \parallel r_{o,\mathrm{PTAIL2}} \right) \approx 43.7 \text{ dB}.$$

The expected total low-frequency gain is therefore

$$\boxed{A_0 \approx A_{v1} + A_{v2} \approx 87 \text{ dB}}.$$

This provides approximately 7 dB of nominal first-order margin relative to the 80 dB design target. The result will subsequently be verified using transistor-level simulation.

4 Ideal Two-Stage Model

An ideal small-signal model is first used to verify the analytical two-stage design independently of MOS transistor parasitics.

Each gain stage is represented by a voltage-controlled current source and its corresponding output resistance. The model parameters are directly obtained from the first-order transistor sizing:

$$g_{m1} = 5\ \mu\text{S}, \qquad R_{\text{out1}} = 29.7\ \text{M}\Omega$$

and

$$g_{m2} = 54.4\ \mu\text{S}, \qquad R_{\text{out2}} = 2.81\ \text{M}\Omega.$$

The two stages are compensated using

$$C_{\text{MILLER}} = 0.8\ \text{pF}.$$

No additional internal capacitance is introduced at this stage. The objective is to isolate the behavior predicted by the first-order two-stage model. The external load capacitance is applied from the testbench.

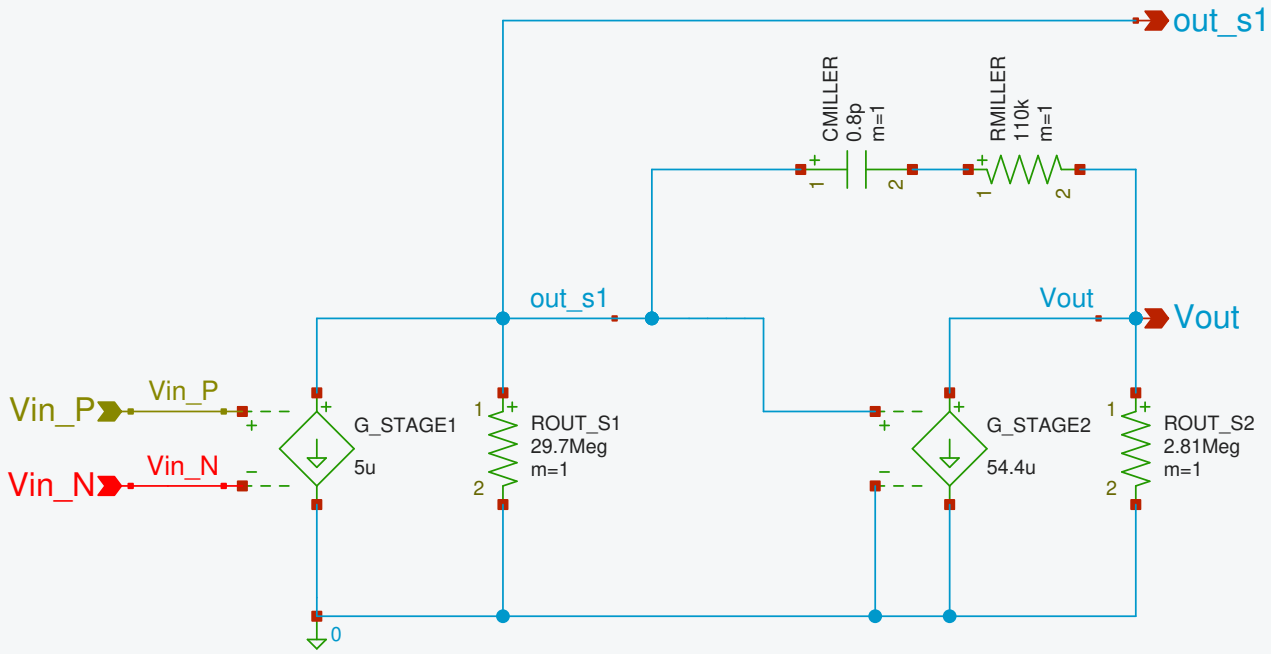


Figure 2 Ideal two-stage OTA model using transconductance stages, calculated output resistances and Miller compensation.

4.1 Nominal AC Response

The initial AC simulation uses

$$C_L = 2\ \text{pF}.$$

The simulated low-frequency gain is

$$A_0 = 87.1 \text{ dB},$$

while the unity-gain frequency is

$$f_{\text{UGF}} = 966.7 \text{ kHz}.$$

The phase at unity gain is

$$\phi(f_{\text{UGF}}) = -107.6^\circ,$$

corresponding to a phase margin of

$$\boxed{PM = 72.4^\circ}.$$

The ideal model therefore reproduces the analytical gain and bandwidth predictions closely while providing comfortable margin relative to the 60° stability target.

4.2 Load-Capacitance Sensitivity

The load capacitance was increased from the nominal value of 2 pF to the specified maximum value of 4 pF, without modifying the OTA parameters.

The resulting AC performance is summarized in Table 3.

Table 3 Ideal-model AC performance versus load capacitance.

C_L	DC gain	UGF	Phase margin
2 pF	87.1 dB	966.7 kHz	72.4°
4 pF	87.1 dB	912.0 kHz	62.5°

As expected, the low-frequency gain is essentially unaffected by the capacitive load. The increased load capacitance lowers the non-dominant pole and therefore reduces the phase margin.

At $C_L = 4$ pF, the model retains approximately 62.5° of phase margin and therefore still satisfies the initial 60° stability target, although with limited additional margin.

4.3 Miller Zero Compensation

Without a series compensation resistor, the Miller feedforward path introduces a right-half-plane zero approximately located at

$$f_{z,\text{RHP}} \approx \frac{g_{m2}}{2\pi C_{\text{MILLER}}} \approx 10.8 \text{ MHz}.$$

For a series Miller resistor, the first-order zero location depends on

$$R_{\text{MILLER}} - \frac{1}{g_{m2}}.$$

For

$$R_{\mathrm{MILLER}} \approx \frac{1}{g_{m2}} \approx 18.4\ \mathrm{k\Omega},$$

the RHP zero is approximately pushed to infinity. At the maximum 4 pF load, this increases the phase margin from 62.5° to 67.4° with negligible bandwidth penalty.

A larger resistance creates a left-half-plane zero. To compensate the phase lag introduced by the non-dominant output pole, the zero was placed approximately around

$$f_{p2} \approx \frac{g_{m2}}{2\pi C_L} \approx 2.2\ \mathrm{MHz}.$$

This gives an initial resistor estimate of approximately

$$R_{\mathrm{MILLER}} \approx 110\ \mathrm{k\Omega}.$$

The resulting LHP zero provides phase lead around the second-pole frequency, making the open-loop response behave approximately as a single-pole system around unity gain without increasing C_{MILLER}.

Table 4 Ideal-model compensation and load study.

C_L	R_{MILLER}	DC gain	UGF	PM
2 pF	0 Ω	87.1 dB	966.7 kHz	72.4°
4 pF	0 Ω	87.1 dB	912.0 kHz	62.5°
4 pF	18.4 kΩ	87.1 dB	909.1 kHz	67.4°
4 pF	110 kΩ	87.1 dB	985.2 kHz	90.1°

The 110 kΩ case demonstrates the intended pole-zero compensation mechanism particularly clearly: the LHP zero approximately counteracts the phase lag of the second pole around the relevant frequency range.

This value is therefore retained as the initial compensation-resistor estimate for the transistor-level design. Its optimum value is expected to shift once transistor parasitics and additional high-frequency poles are introduced.

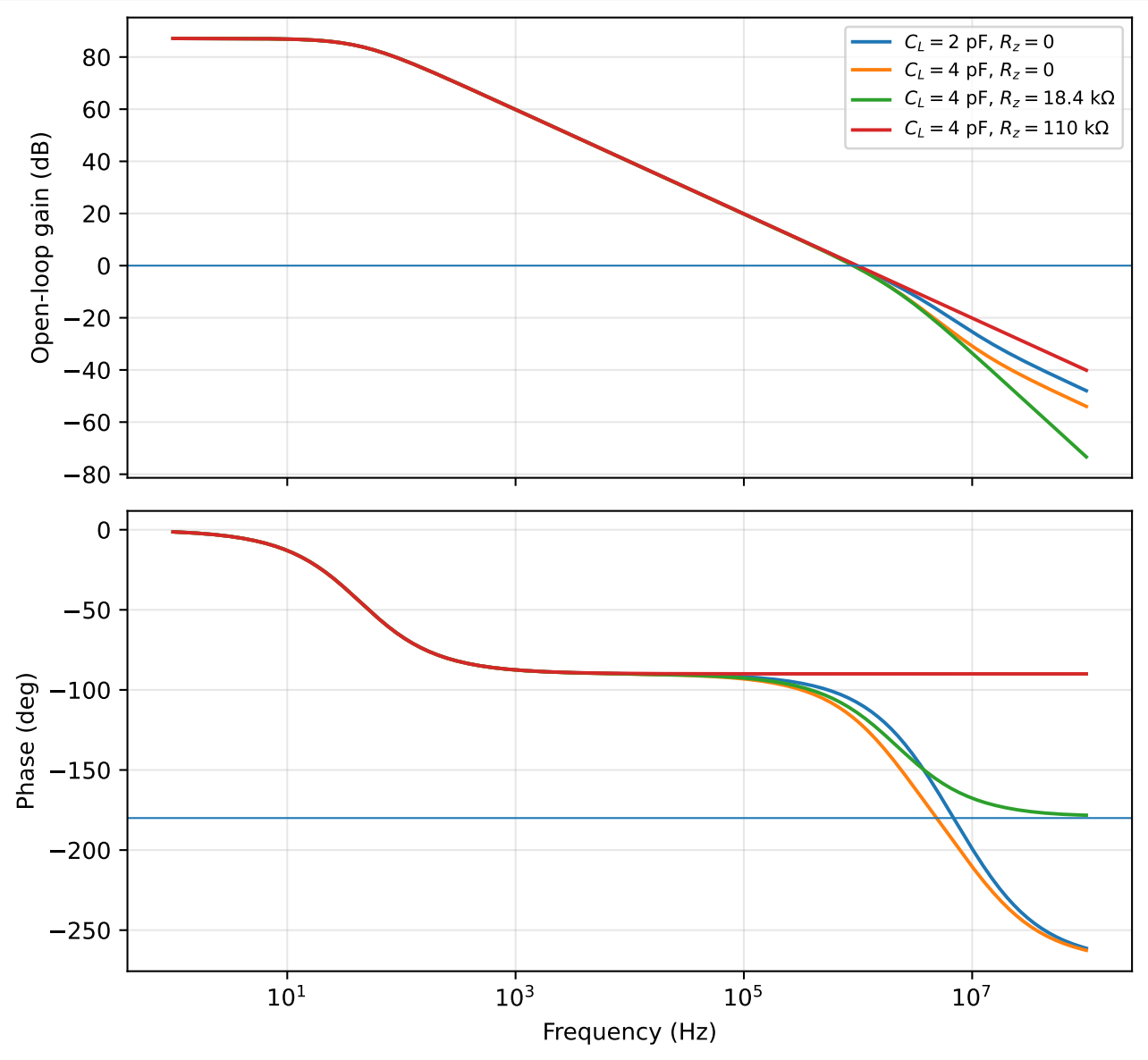


Figure 3 Ideal open-loop response showing the effect of load capacitance and Miller zero compensation.

5 Transistor-Level Design

The ideal two-stage model is replaced by the complete SKY130A transistor-level implementation. Device dimensions are first obtained from the g_m/I_D characterization, then refined using DC, AC, and statistical simulations. The sequence is important: the first working transistor-level circuit is a *starting point*, not the final sizing used for characterization in Section 6.

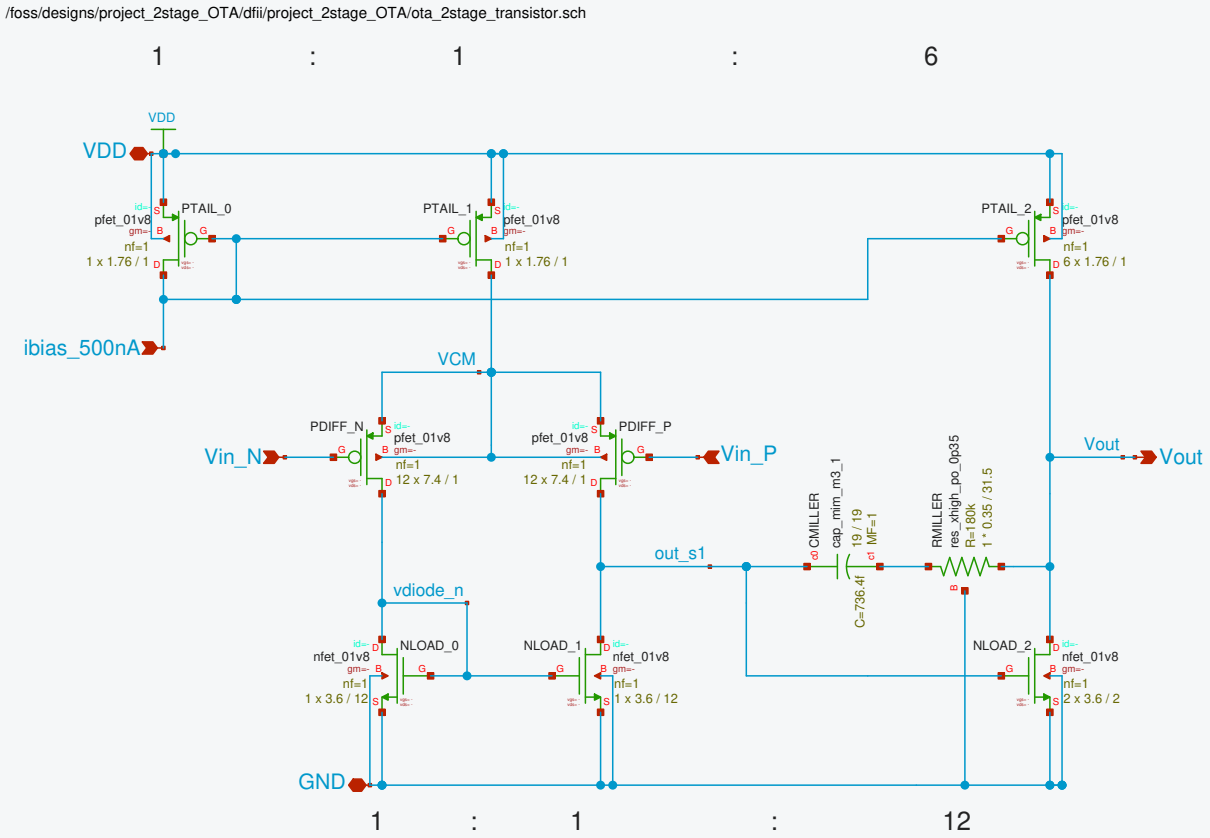


Figure 4 Final transistor-level implementation of the two-stage OTA using SKY130A devices.

5.1 Unit-Device Implementation

The first-order sizing supplies effective transistor widths. Larger transistors are represented using parallel unit devices, allowing the bias ratios to be expressed directly through device multiplicity. With reference current $I = 500$ nA, the intended PMOS current-source ratio is

$$\boxed{\texttt{PTAIL_0} : \texttt{PTAIL_1} : \texttt{PTAIL_2} = 1 : 1 : 6},$$

so that the first-stage tail current is approximately I and the second-stage bias current is approximately $6I = 3\ \mu$A. At zero differential input, the first-stage load devices each carry approximately $I/2$.

The original NMOS implementation used identical unit geometries, with multiplicities $1 : 1 : 12$. This follows from the branch-current ratio

$$\frac{I_{\text{second stage}}}{I_{\text{first branch}}} = \frac{6I}{I/2} = 12.$$

The final implementation retains a *geometric effective-W/L ratio* of $1 : 1 : 12$ but does *not* use identical NMOS units. Instead, the first-stage mirror is enlarged for mismatch performance, while the second-stage device is kept comparatively compact to limit loading of the high-impedance node `out_s1`.

Because the channel lengths differ, an equal effective-W/L scaling ratio is not by itself proof of identical current density or precise current mirroring; those depend on the full PDK models and bias points.

5.2 Final Transistor Dimensions

Table 5 lists the geometries retained after offset and stability optimization. Widths and lengths are given in micrometers; m denotes parallel-device multiplicity.

Table 5 Selected final transistor sizing. Effective width is mW and is not a replacement for the explicit (W, L, m) used in the PDK netlist.

Device	Unit W/L (μm/μm)	m	Effective W (μm)
PTAIL_0	1.76 / 1	1	1.76
PTAIL_1	1.76 / 1	1	1.76
PTAIL_2	1.76 / 1	6	10.56
PDIFF_N	7.4 / 1	12	88.8
PDIFF_P	7.4 / 1	12	88.8
NLOAD_0	3.6 / 12	1	3.6
NLOAD_1	3.6 / 12	1	3.6
NLOAD_2	3.6 / 2	2	7.2

5.2.1 Initial NMOS Unit-Length Refinement

The first transistor-level iterations used an NMOS unit with $W/L = 0.45/1$ μm/μm. The resulting low-frequency open-loop gain was approximately 79.2 dB, below the 80 dB target. Increasing L alone while holding W fixed changed the inversion level and did not improve gain. Resizing both dimensions to $0.60/2$ μm/μm restored a useful transconductance efficiency and output resistance. For this *intermediate* sizing, the simulated first-stage NMOS loads gave approximately

$$\frac{g_m}{I_D} \approx 17.65\ \mathrm{V}^{-1}, \qquad \frac{g_m}{g_{ds}} \approx 134,$$

while the second-stage NMOS gave approximately

$$\frac{g_m}{I_D} \approx 17.54\ \mathrm{V}^{-1}, \qquad \frac{g_m}{g_{ds}} \approx 133.$$

These are historical operating-point values and should not be attributed to the final, independently resized NMOS devices.

5.2.2 Offset-Driven Mirror Resizing

The 0.60/2 first-stage mirror was adequate for nominal AC behavior but suffered significant random input-referred offset. At $V_{DD} = 1.8$ V and $T = 27°$C, the initial 300-sample Monte Carlo analysis yielded

$$\sigma(V_{OS}) \approx 6.125\ \mathrm{mV}.$$

The two NLOAD_0/1 devices were therefore enlarged by scaling W and L together, preserving their geometric ratio $W/L = 0.3$. The first change, from 0.60/2 to 2.4/8, multiplied the gate area by 16 and reduced offset dispersion to approximately 1.448 mV. Subsequent increases gave diminishing returns; the selected compromise is

$$\boxed{\texttt{NLOAD_0/1}: \quad W = 3.6\ \mu\mathrm{m}, \quad L = 12\ \mu\mathrm{m}, \quad m = 1.}$$

The complete statistical iteration history is reported in Table 10.

5.2.3 Second-Stage NMOS Parasitic Trade-off

The second-stage transistor is a functional, active common-source device, not a layout dummy. An intermediate large-area choice ($W = 3.6\ \mu$m, $L = 6\ \mu$m, $m = 6$) preserved the intended effective-W/L scaling, but introduced substantial gate capacitance at `out_s1` and degraded phase margin.

The retained device is

$$\boxed{\texttt{NLOAD_2}: \quad W = 3.6\ \mu\text{m}, \quad L = 2\ \mu\text{m}, \quad m = 2.}$$

Its geometric scaling relative to either first-stage NMOS mirror device is

$$\frac{m_2(W/L)_2}{m_{0,1}(W/L)_{0,1}} = \frac{2(3.6/2)}{1(3.6/12)} = 12.$$

The retained gate area, $mWL = 14.4\ \mu\text{m}^2$, is one-ninth of the large-area intermediate choice, $129.6\ \mu\text{m}^2$. Gate area is only a rough guide to capacitance; device terminal capacitances are evaluated by the transistor models in AC simulation. At the earlier $C_M \approx 800$ fF, $R_M = 110\ \text{k}\Omega$ comparison point with $C_L = 4$ pF, this change recovered the nominal phase margin from approximately 49.9° to 67.7°, without materially changing the offset dispersion.

5.3 DC Operating Point Verification

The biasing concept is evaluated with the OTA in unity-gain feedback, using

$$V_{DD} = 1.8\ \text{V}, \qquad V_{\text{in}} = 0.6\ \text{V}, \qquad I_{\text{BIAS}} = 500\ \text{nA}.$$

For the *earlier* 0.60/2 *NMOS-unit implementation*, an operating-point check returned the following DC node voltages:

Node	Earlier sizing
V_{out}	0.600018 V
$V_{\text{out_s1}}$	0.633 V
$V_{\text{diode,n}}$	0.631 V
V_{CM} (pair common-source node)	1.402 V
$V_{\text{bias,p}}$	0.765 V

The associated closed-loop tracking error was approximately $-18\ \mu$V; this was a finite-loop-gain DC check, not an input-referred mismatch-offset measurement.

These internal-node voltages are *not* reused for the final operating point. Both the first-stage NMOS mirror and the second-stage NMOS were subsequently resized. The completed TT PVT test of the selected final circuit at 1.8 V and 27°C reports a total supply current of

$$\boxed{I_Q = 4.001\ \mu\text{A}.}$$

A new final-device DC operating-point extraction would be required to report updated g_m, g_{ds}, and internal-node voltages or to establish numerical common-mode headroom limits; these quantities are therefore not inferred from the older data.

5.4 Small-Signal Gain: Estimate and Final Verification

For a first-order two-stage approximation, the unloaded stage gains are estimated from the transconductances and output conductances:

$$A_{v1} \approx \frac{g_{m,\text{PDIFF}}}{g_{ds,\text{PDIFF}} + g_{ds,\text{NLOAD1}}}, \qquad A_{v2} \approx \frac{g_{m,\text{NLOAD2}}}{g_{ds,\text{NLOAD2}} + g_{ds,\text{PTAIL2}}}.$$

For the *earlier* 0.60/2 NMOS-unit design, the operating-point values gave approximately 41.0 dB and 40.8 dB for the two stages, corresponding to $A_0 \approx 81.8$ dB. Those estimates explain the initial viability of the topology, but they are not the final stage gains.

With the selected mirror sizing and compensated transistor-level circuit, the low-frequency gain is instead measured directly by AC simulation:

$$\boxed{A_{0,\text{final}} = 89.73 \text{ dB}.}$$

The larger first-stage mirror devices modify the output conductance and the high-impedance internal node, while the changed second-stage geometry also modifies the small-signal operating conditions. Accordingly, a final per-stage analytical gain breakdown would require new operating-point g_m and g_{ds} values rather than simply carrying forward those of the previous sizing.

5.5 Physical Implementation of the Compensation Network

The compensation mechanism is initially investigated with ideal passive components, then realized using physical SKY130A devices. The analytical 0.8 pF / 110 kΩ solution is retained as a *starting design*; the final values are determined by transistor-level AC and PVT stability checks.

5.5.1 Miller Capacitor Selection

A metal–insulator–metal capacitor is preferred over a strongly voltage-dependent MOS structure for a nominally fixed compensation capacitance. The selected PDK primitive is

`sky130_fd_pr__cap_mim_m3_1`

For this device, the Xschem symbol computes the nominal capacitance from its dimensions as

$$C_{\text{nom}} = MF\,[2WL + 0.38(W + L)] \text{ fF},$$

where W and L are in micrometers. The initial $W = L = 19.81$ μm gave approximately 799.93 fF. After retuning, the selected geometry is

$$W = L = 19 \ \mu\text{m}, \qquad MF = 1,$$

so that

$$\boxed{C_{\text{MILLER,nom}} = 736.44 \text{ fF}.}$$

The complete PDK capacitor model remains in use for the simulated circuit, including its device parasitics.

5.5.2 Miller Resistor Selection

The compensation resistor uses the physical high-resistance poly primitive

`sky130_fd_pr__res_xhigh_po_0p35`

Using the nominal sheet resistance $R_\square \approx 2$ kΩ/□ gives the sizing estimate

$$R_{\text{nom}} = R_\square \frac{L}{Wm}.$$

The original design used $W = 0.35$ μm and $L = 19.25$ μm for 110 kΩ. The final PVT study favored a larger resistance, implemented with

$$W = 0.35 \ \mu\text{m}, \qquad L = 31.5 \ \mu\text{m}, \qquad m = 1,$$

which gives

$$\boxed{R_{\mathrm{MILLER,nom}} = 180\ \mathrm{k\Omega}.}$$

This large series resistance establishes a left-half-plane compensation zero, providing useful phase lead near the relevant non-dominant-pole region. The final compensation branch uses physical SKY130A capacitor and resistor primitives, rather than idealized SPICE elements.

5.6 Final Open-Loop AC Verification

The selected transistor geometry and physical compensation network are

$$\boxed{C_{\mathrm{MILLER}} = 736.44\ \mathrm{fF}, \qquad R_{\mathrm{MILLER}} = 180\ \mathrm{k\Omega}.}$$

At the maximum specified load, $C_L = 4$ pF, the final AC regression gives

$$\boxed{A_0 = 89.7\ \mathrm{dB}, \qquad f_{\mathrm{UGF}} = 1068\ \mathrm{kHz}, \qquad PM = 75.0^\circ.}$$

At nominal load, $C_L = 2$ pF, the corresponding unity-gain frequency is 1184 kHz and the phase margin is 89.0°. The 1 MHz nominal-bandwidth target is achieved, while both loads retain phase margin greater than 60°.

Table 6 Final nominal AC performance for the selected physical compensation network.

C_L	DC gain	UGF	Phase margin
2 pF	89.7 dB	1184 kHz	89.0°
4 pF	89.7 dB	1068 kHz	75.0°

The full 45-condition PVT regression at $C_L = 4$ pF reports a minimum gain of 87.63 dB and minimum phase margin of 64.69°. These results are developed in Section 6 and verify that the selected nominal compensation is not relying solely on a favorable TT operating point.

5.7 Miller Compensation at Transistor Level

The role of the series resistor is verified at $C_L = 4$ pF using four physical or reference configurations:

$$R_{\mathrm{MILLER}} \in \{0,\ 19,\ 110,\ 180\}\ \mathrm{k\Omega},$$

A near-short series path retains the original right-half-plane feedforward zero. Around $1/g_{m2} \approx 19\ \mathrm{k\Omega}$, the zero is approximately nulled. Larger resistances create a left-half-plane zero and add phase lead around the crossover region.

Table 7 Final transistor-level compensation study at $C_L = 4$ pF.

R_{MILLER}	DC gain	UGF	Phase margin
Short reference	89.7 dB	863.2 kHz	42.5°
19 kΩ physical	89.7 dB	863.9 kHz	46.8°
110 kΩ physical	89.7 dB	925.4 kHz	64.7°
180 kΩ physical (selected)	89.7 dB	1067.6 kHz	75.0°

With the final 736.44 fF MIM capacitor, the nominal 4-pF phase margin improves from approximately 64.65° at 110 kΩ to 74.98° at 180 kΩ. A PVT check with 110 kΩ had produced a worst-case phase margin of 55.23°, below the specified 60° minimum. Retuning to 180 kΩ recovered a worst-case phase margin of 64.69° and maintained a minimum gain margin of 13.76 dB across the same 45 PVT conditions.

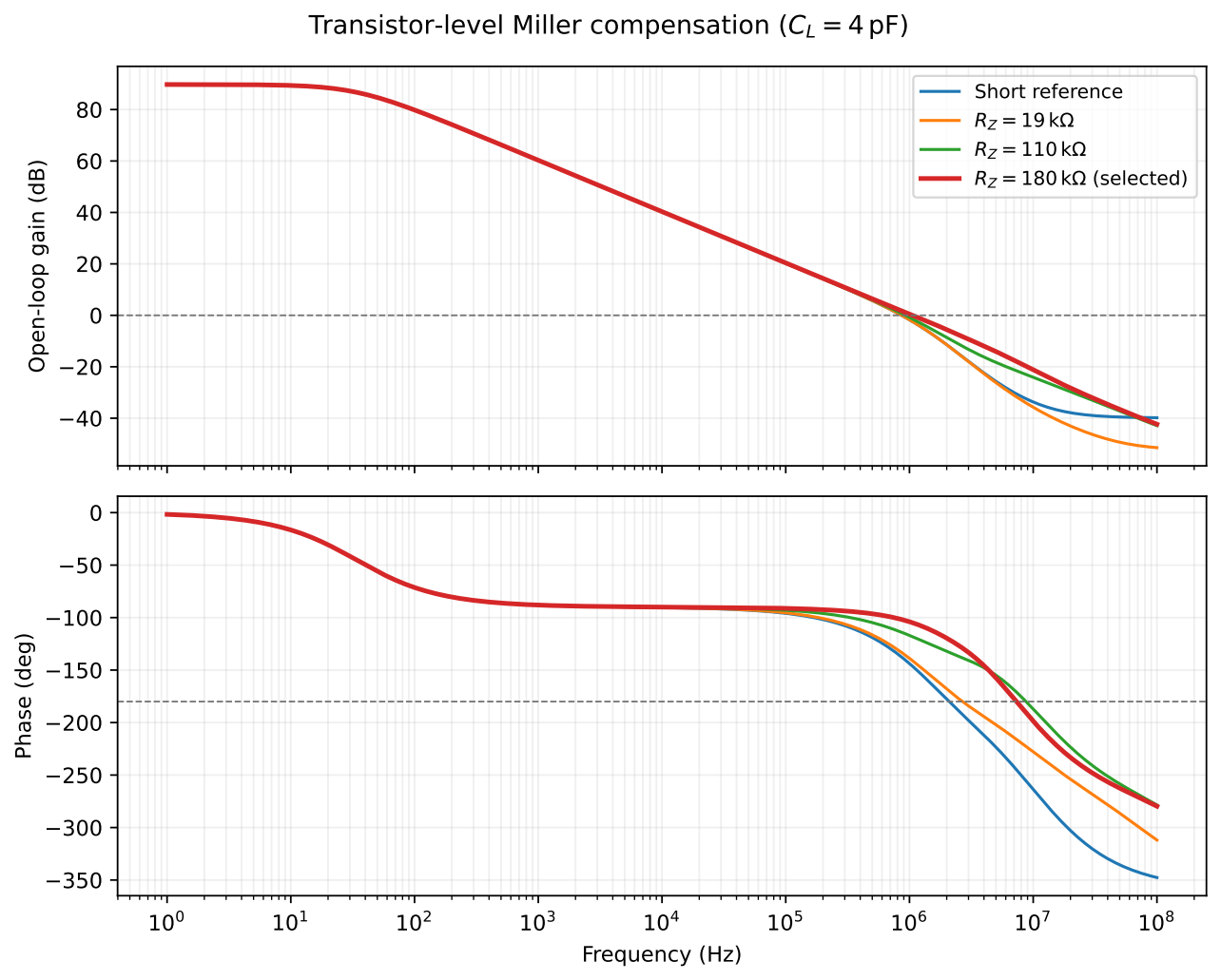


Figure 5 Transistor-level open-loop response versus Miller resistance at $C_L = 4$ pF.

Therefore, 180 kΩ is the *selected final* series-resistor value: it is justified by both the physical AC comparison and the full PVT stability requirement, rather than only by the idealized two-stage analysis.

6 Characterization

6.1 Slew-Rate Verification

The final OTA is configured as a unity-gain follower and driven by a 0.45–0.75 V input step with $C_L = 4$ pF. The slew rate is estimated from the average 20–80% output-transition slope, with the local derivative around the transition midpoint used as an additional check.

The simulated 20–80% slopes are

$$\boxed{SR_{\text{rise}} = 0.549\ \text{V}/\mu\text{s}}, \qquad \boxed{SR_{\text{fall}} = 0.469\ \text{V}/\mu\text{s}}.$$

The corresponding midpoint derivatives are

$$SR_{\text{rise,mid}} = +0.558\ \text{V}/\mu\text{s}, \qquad SR_{\text{fall,mid}} = -0.429\ \text{V}/\mu\text{s}.$$

The two measurements reflect slightly different regions of each transition; in particular, the falling edge is not perfectly linear. The 20–80% values are retained as the reported slew rates.

A first-order first-stage current-limited estimate gives

$$\frac{I_{\text{tail,1}}}{C_M} \approx \frac{0.5\ \mu\text{A}}{736.44\ \text{fF}} \approx 0.68\ \text{V}/\mu\text{s}.$$

The simulated large-signal response also depends on current redistribution, second-stage dynamics, the output load, and transistor parasitics; this simple estimate is therefore not an exact prediction.

6.2 Saturation and Headroom Criterion

The usable input and output voltage ranges are assessed using both compact-model operating-point information and a practical source-to-drain headroom criterion, rather than relying exclusively on an inversion-region flag.

At 27°C, the thermal voltage is

$$U_T = \frac{kT}{q} \approx 25.9\ \text{mV}.$$

For this first-order headroom assessment, a conservative screening criterion is chosen as

$$\boxed{|V_{DS}| \gtrsim 4U_T \approx 0.10\ \text{V}}.$$

This is an engineering approximation for the present low-current design, not a universal saturation condition for every MOS operating point. Final usable ranges must also be checked against bias-current regulation, gain, and the full compact-model behavior.

6.3 Input Common-Mode Range

The first stage uses the PMOS input pair `PDIFF_N/P`, biased by `PTAIL_1` and loaded by the diode-connected NMOS mirror `NLOAD_0/1`.

6.3.1 Lower ICMR Limit

The PMOS pair requires sufficient source-to-drain headroom,

$$V_{SD,\text{PDIFF}} \gtrsim 4U_T.$$

With

$$V_{S,\mathrm{PDIFF}} = V_{CM} + V_{SG,\mathrm{PDIFF}}$$

and the diode-connected load approximately fixing one input-device drain voltage at $V_{GS,\mathrm{NLOAD0}}$, the lower limit is estimated as

$$V_{CM,\mathrm{min}} \approx V_{GS,\mathrm{NLOAD0}} + 4U_T - V_{SG,\mathrm{PDIFF}}.$$

For single-supply operation, the practical range is additionally bounded by ground:

$$V_{CM,\mathrm{min,usable}} \gtrsim 0\ \mathrm{V}.$$

6.3.2 Upper ICMR Limit

At large V_{CM}, the first-stage PMOS source device `PTAIL_1` becomes limiting:

$$V_{SD,\mathrm{PTAIL1}} = V_{DD} - (V_{CM} + V_{SG,\mathrm{PDIFF}}) \gtrsim 4U_T.$$

Thus,

$$\boxed{V_{CM,\mathrm{max}} \approx V_{DD} - V_{SG,\mathrm{PDIFF}} - 4U_T}.$$

The earlier numerical ICMR estimate was based on operating-point voltages from the initial transistor sizing. Since `NLOAD_0/1` were substantially resized, those voltages have not been reused here. A numerical ICMR specification for the final sizing requires an updated operating-point extraction and common-mode sweep; the equations above are topology-based design estimates only.

6.4 Output Common-Mode Range

The second stage comprises the PMOS current source `PTAIL_2` and the active NMOS common-source device `NLOAD_2`. Assuming the same first-order $4U_T$ headroom criterion at each output device gives

$$V_{\mathrm{out,min}} \approx 4U_T \approx 0.10\ \mathrm{V},$$

$$V_{\mathrm{out,max}} \approx V_{DD} - 4U_T \approx 1.70\ \mathrm{V} \quad (V_{DD} = 1.8\ \mathrm{V}).$$

The resulting topology-based nominal estimate is

$$\boxed{\mathrm{OCMR}_{\mathrm{est.}} \approx 0.10\text{–}1.70\ \mathrm{V}}.$$

This is a headroom estimate, not a final measured output-swing specification. A dedicated large-signal sweep on the final device sizing is required to verify current regulation, distortion, and usable output limits.

6.5 Positive-Supply Rejection

The final OTA is simulated as a unity-gain follower around a 0.6 V reference, with $V_{DD} = 1.8$ V and a normalized small-signal supply excitation of 1 V. The positive-supply rejection ratio is calculated as

$$PSRR_+(f) = 20\log_{10}\left|\frac{v_{DD}(f)}{v_{\mathrm{out}}(f)}\right|.$$

At low frequency, the final nominal result is approximately

$$\boxed{PSRR_+(1\ \mathrm{kHz}) = 99.86\ \mathrm{dB}}.$$

As frequency increases, the declining feedback gain and parasitic supply-to-output coupling reduce rejection. The load-dependent 2-pF/4-pF comparison from the original sizing is intentionally not reproduced here, as it was not rerun for the final circuit.

Table 8 Final nominal positive-supply rejection from the regenerated PSRR testbench.

Frequency	$PSRR_+$
1 Hz	99.88 dB
1 kHz	99.86 dB
100 kHz	84.73 dB
1 MHz	59.02 dB
10 MHz	49.85 dB

6.6 Noise

The final OTA is characterized as a unity-gain follower at $V_{DD} = 1.8$ V, with the input reference at 0.6 V and $C_L = 2$ pF. The integrated noise interval is

$$1 \text{ Hz} \leq f \leq 1 \text{ MHz}.$$

The final integrated results are

$$\boxed{v_{n,\text{out,rms}} = 120.27\ \mu\text{V}}, \qquad \boxed{v_{n,\text{in,rms}} = 134.66\ \mu\text{V}}.$$

Compared with the initial sizing (175.53 μV output RMS and 191.09 μV input RMS), these represent reductions of approximately 31.5% and 29.5%, respectively.

Device-level integration identifies the first-stage NMOS mirror as the largest output-noise contributor. Squaring each device's integrated RMS contribution and dividing by the squared total output noise gives

$$\underbrace{38.0\%}_{\texttt{NLOAD_1}} + \underbrace{34.8\%}_{\texttt{NLOAD_0}} \approx 72.8\%.$$

The two PMOS input devices jointly contribute approximately 20.3% of the output-noise power; the compensation resistor contributes about 3.8% and `NLOAD_2` approximately 1.5%. Other devices account for the remainder.

The load-mirror contribution has decreased from approximately 87.6% in the original circuit to 72.8% with the selected larger NMOS mirror devices. The mirror remains the dominant noise source, but the PMOS input pair is no longer negligible. In this final simulation the integrated NMOS mirror noise contains a substantial channel thermal component, explaining why additional area increases need not produce a proportional reduction in total integrated noise.

The noise ranking separately reports device totals and their internal mechanisms (e.g. channel and $1/f$ noise); those entries must not be summed together as independent device contributions.

6.7 PVT Analysis

The final transistor-level OTA was evaluated over five process corners: SS, SNFP (slow NMOS/fast PMOS), TT, FNSP (fast NMOS/slow PMOS), and FF. Each was combined with

$$V_{DD} \in \{1.62,\ 1.80,\ 1.98\}\ \text{V} \quad \text{and} \quad T \in \{-40,\ 27,\ 125\}\ ^\circ\text{C},$$

for 45 operating conditions in total. The sweep used the maximum specified load, $C_L = 4$ pF, and the selected physical compensation network,

$$C_M = 736.44\ \text{fF}, \qquad R_M = 180\ \text{k}\Omega.$$

The reported metrics are low-frequency open-loop gain, unity-gain frequency, phase margin, gain margin, quiescent current, and PSRR at 1 kHz.

At TT, $V_{DD} = 1.8$ V and $T = 27°$C, the 4-pF results are

$$A_0 = 89.73 \text{ dB}, \qquad f_{\text{UGF}} = 1.0676 \text{ MHz},$$

$$PM = 74.98°, \qquad GM = 17.9 \text{ dB},$$

with $I_Q = 4.001$ μA and $PSRR_+(1 \text{ kHz}) = 99.86$ dB.

Table 9 Main electrical performance over the 45-point PVT matrix at $C_L = 4$ pF.

Metric	Nominal	Worst PVT	Worst condition
DC gain	89.7 dB	87.6 dB	FNSP / +125°C / 1.62 V
UGF	1068 kHz	828 kHz	FNSP / +125°C / 1.62 V
Phase margin	75.0°	64.7°	SS / +125°C / 1.62 V
Gain margin	17.9 dB	13.8 dB	SNFP / −40°C / 1.98 V
Supply current	4.00 μA	4.05 μA	FNSP / +125°C / 1.98 V
PSRR$^+$ @ 1 kHz	99.9 dB	53.3 dB	SNFP / −40°C / 1.62 V

The minimum DC gain is 87.63 dB (FNSP, 125°C, 1.62 V). The minimum phase margin is 64.69° (SS, 125°C, 1.62 V), while the minimum gain margin is 13.76 dB (SNFP, −40°C, 1.98 V). Thus, all 45 cases meet the hard criteria

$$\boxed{A_0 \geq 80 \text{ dB}, \quad PM \geq 60°, \quad GM > 10 \text{ dB}}.$$

The UGF ranges from 827.8 kHz to 1.4652 MHz across PVT at $C_L = 4$ pF. The nominal-bandwidth target of 1 MHz was defined for $C_L = 2$ pF, for which the final TT simulation reaches 1.1838 MHz.

The supply-current range is 3.787–4.050 μA, and the lowest simulated PSRR at 1 kHz is 53.28 dB (SNFP, −40°C, 1.62 V). This significant corner sensitivity is reported separately from the gain and stability requirements rather than being described as invariant.

6.8 Monte Carlo Analysis

Monte Carlo analysis was used to quantify the effects of the SKY130 statistical process and local-mismatch models on the input-referred offset V_{OS} and low-frequency open-loop gain A_0. Both variation mechanisms were enabled. The final campaign used 300 independent samples per combination of

$$V_{DD} \in \{1.62,\ 1.80,\ 1.98\} \text{ V}, \qquad T \in \{-40,\ 27,\ 125\}\ ^\circ\text{C},$$

for a total of 2700 simulations, at $C_L = 4$ pF. The offset is input-referred using the unity-follower DC solution, with a finite-gain correction using the sampled A_0; gain is extracted from the low-frequency AC response of the same randomized sample. These results describe the available open-source statistical models and are not claimed to be foundry-qualified sign-off data.

6.8.1 Offset-Driven Sizing Iterations

The initial input-referred offset standard deviation at 1.8 V and 27°C was 6.125 mV. This motivated an iterative increase of the first-stage NMOS mirror-load area while approximately preserving its unit-device W/L ratio. The two load transistors were initially much smaller in gate area than the PMOS input pair.

Increasing the mirror-load area initially reduced the offset standard deviation by a factor of approximately four, broadly consistent with the first-order matching trend

$$\sigma_{\text{mismatch}} \propto \frac{1}{\sqrt{WL}}.$$

Table 10 Offset optimization at $V_{DD} = 1.8$ V and $T = 27°C$ ($N = 300$ for each design). Transistor W/L is given in $\mu m/\mu m$; the final row uses the retuned Miller network.

Design	NLOAD_0/1	NLOAD_2	$\sigma(V_{OS})$	$3\sigma(V_{OS})$
Initial	0.6/2, $m = 1$	0.6/2, $m = 12$	6.125 mV	18.37 mV
Iter. 1	2.4/8, $m = 1$	0.6/2, $m = 12$	1.448 mV	4.34 mV
Iter. 2	3.6/12, $m = 1$	3.6/6, $m = 6$	1.195 mV	3.58 mV
Iter. 3	3.6/12, $m = 1$	3.6/2, $m = 2$	1.163 mV	3.49 mV
Iter. 4	4.2/14, $m = 1$	3.6/2, $m = 2$	1.177 mV	3.53 mV
Final	3.6/12, $m = 1$	3.6/2, $m = 2$	1.109 mV	3.33 mV

Changing NLOAD_2 separately recovered stability by limiting the gate capacitance at the high-impedance first-stage output node out_s1. This device is an *active* second-stage NMOS, not a non-functional layout dummy. In the selected sizing,

$$\frac{m_2(W/L)_2}{m_{0,1}(W/L)_{0,1}} = \frac{2(3.6/2)}{1(3.6/12)} = 12.$$

This is an effective geometric W/L ratio, not a guarantee of an identical current-density relationship between devices of different channel length.

The subsequent Iteration 4 produced no statistically significant reduction in offset dispersion and required more mirror area. Iteration 3 was therefore retained as the transistor-sizing baseline; Miller-compensation optimization subsequently produced the selected final circuit. The small difference between the Iteration-3 and final Monte Carlo estimates should not be ascribed solely to the compensation change because the results come from independent 300-sample datasets.

6.8.2 Final Statistical Performance

Table 11 Monte Carlo summary for process and mismatch variations.

V_{DD} [V]	T [°C]	$\mu(V_{OS})$ [mV]	$\sigma(V_{OS})$ [mV]	$\mu(A_0)$ [dB]	$\sigma(A_0)$ [dB]
1.62	-40	+0.31	1.24	89.04	0.129
1.80	-40	+0.25	1.24	89.46	0.180
1.98	-40	+0.30	1.25	89.62	0.180
1.62	27	+0.20	1.28	89.83	0.073
1.80	27	+0.04	1.11	90.04	0.070
1.98	27	+0.19	1.15	90.21	0.080
1.62	125	+0.10	1.14	89.31	0.054
1.80	125	+0.11	1.15	89.56	0.061
1.98	125	+0.13	1.20	89.76	0.064

At nominal supply and temperature, the final offset distribution has

$$\boxed{\mu(V_{OS}) = +0.0396 \text{ mV}, \qquad \sigma(V_{OS}) = 1.1089 \text{ mV}},$$

so $3\sigma(V_{OS}) = 3.327$ mV. This is an 81.9% reduction in offset standard deviation relative to the initial design. The final mean offset is small compared with the spread and the distribution must not be summarized by its mean alone.

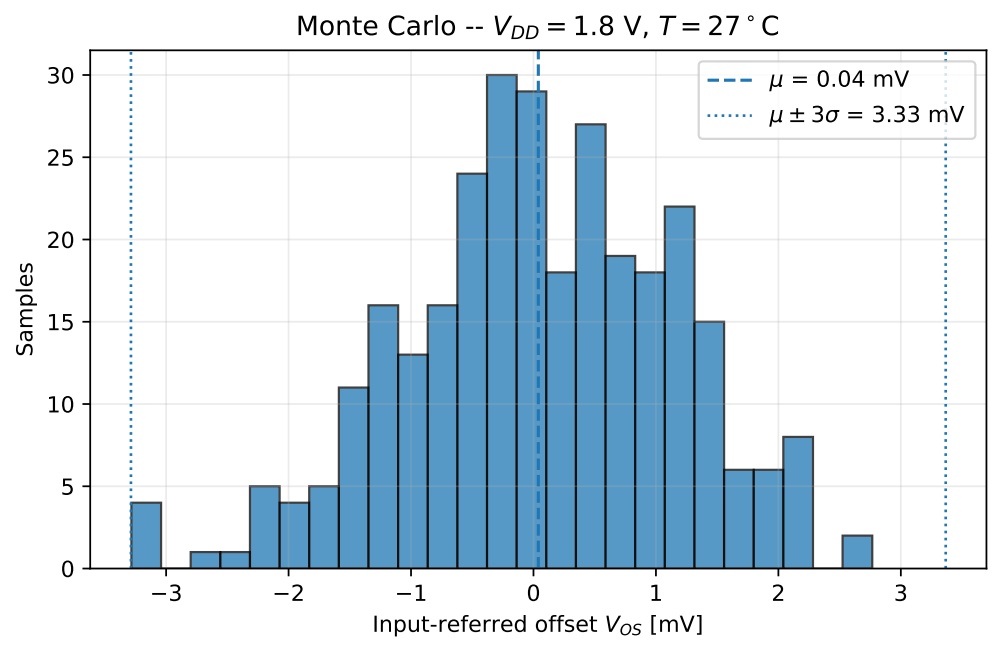


Figure 6 Final input-referred offset distribution at $V_{DD} = 1.8$ V and $T = 27°C$ ($N = 300$).

The corresponding nominal gain distribution has

$$\boxed{\mu(A_0) = 90.04 \text{ dB}, \qquad \sigma(A_0) = 0.070 \text{ dB}}.$$

The nominal lower $\mu - 3\sigma$ estimate is approximately 89.83 dB, well above the 80 dB target. The deterministic TT gain of 89.73 dB need not coincide exactly with the mean of the randomized ensemble.

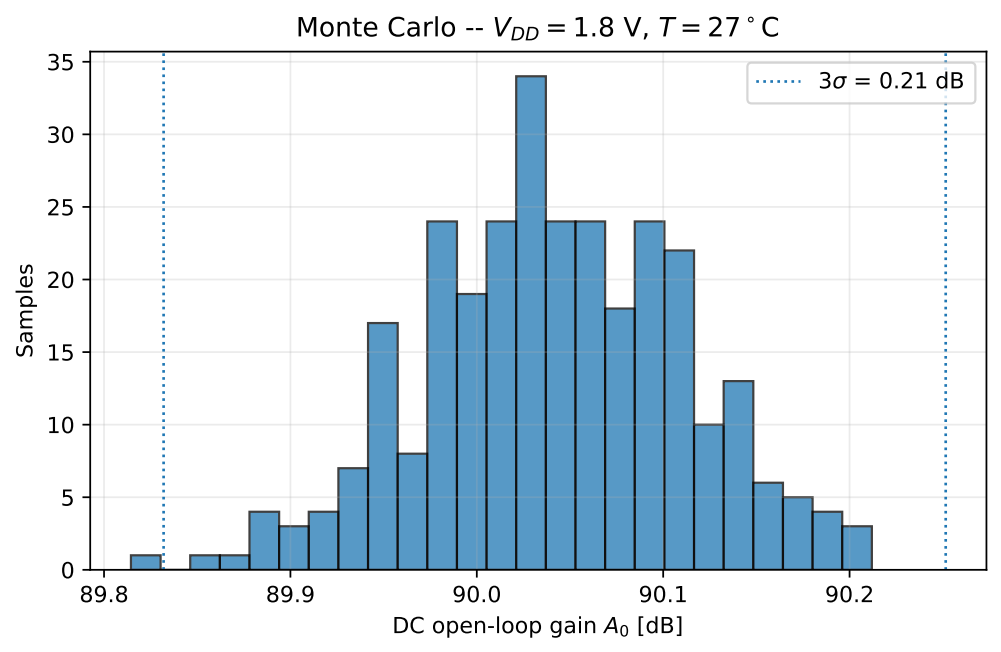


Figure 7 Final low-frequency open-loop gain distribution at $V_{DD} = 1.8$ V and $T = 27°C$ ($N = 300$).

Across the nine conditions, the maximum observed offset standard deviation is 1.277 mV, corresponding to $3\sigma = 3.831$ mV at $V_{DD} = 1.62$ V and $T = 27°C$. The largest absolute mean offset is 0.313 mV at 1.62 V and $-40°C$. For a two-sided absolute-offset estimate, the worst value of $|\mu| + 3\sigma$ across these conditions is approximately 4.06 mV at 1.98 V and $-40°C$.

The minimum gain-ensemble mean is 89.04 dB and the most conservative lower $\mu - 3\sigma$ estimate among the nine conditions is approximately 88.65 dB, above the gain requirement. The optimization objective of approximately 3 mV at 3σ was not fully met (even at nominal conditions), despite the substantial improvement. This trade-off is reported explicitly instead of treating a statistical target as achieved.

7 Conclusion

This project developed a low-power two-stage CMOS OTA using the SKY130A open-source PDK. The design progressed from first-order calculations and an ideal two-stage model to a transistor-level implementation. This approach made it possible to study the gain, pole–zero behavior, and Miller compensation independently before introducing device limitations and parasitic effects.

The final OTA operates from a nominal 1.8 V supply and draws approximately 4.0 μA, including the reference bias branch. With the physical compensation network $C_{\mathrm{MILLER}} = 736.44$ fF and $R_{\mathrm{MILLER}} = 180$ kΩ, the nominal open-loop DC gain is 89.73 dB. At the specified 2 pF nominal load, the unity-gain frequency reaches 1.184 MHz with a phase margin of 89.04°. At the maximum 4 pF load, the phase margin remains 74.98°. All 45 simulated PVT conditions satisfy the hard gain and stability requirements, with minimum values of 87.63 dB DC gain, 64.69° phase margin, and 13.76 dB gain margin.

Statistical characterization also informed the design choices. Increasing the area of the first-stage NMOS mirror loads reduced the nominal input-referred offset standard deviation from 6.125 mV to 1.109 mV. Subsequent iterations showed diminishing returns, while reducing the capacitive loading introduced by the second-stage NMOS helped recover stability. The final nominal offset spread is $3\sigma(V_{OS}) = 3.33$ mV; the largest value among the nine voltage–temperature conditions is 3.83 mV. Thus, the approximately 3 mV statistical-offset objective was not fully met, despite the substantial improvement. The integrated output noise was also reduced to 120.27 $\mu\mathrm{V}_{\mathrm{RMS}}$ over 1 Hz–1 MHz.

The remaining limitations are primarily related to physical implementation and verification. The reported results are schematic-level simulations using the available SKY130A models; no layout, extracted parasitics, or silicon measurements are included. A natural continuation would be a matching-aware layout followed by DRC, LVS, parasitic extraction, and post-layout characterization. This would allow the stability, offset, noise, and usable input/output voltage ranges to be reassessed with physical parasitics and layout-dependent effects.